Aleksandr Medvedev

Evaluation | RAG | PyTorch | TensorFlow

30+ h/week | medv.ax@outlook.com | github.com/MedvAx-AI | medvax-ai.github.io

SUMMARY

3rd-year Applied Artificial Intelligence student focused on empirical LLM research: evaluation, efficiency, RAG, reproducibility, and safety controls. Designed a Qwen2.5 routing study across 1,000 runs on 500 MMLU questions, identifying 9.8 pp of routing headroom and reducing p50 latency by 15%. Built leakage-free held-out evaluation with bootstrap confidence intervals and 1,000/1,000 reproducible rerun answers. Interested in LLM architecture, training methods, model quality, efficiency, safety, and risk analysis.

SKILLS

LLM Research: LLM evaluation, RAG, Hugging Face Transformers, Qwen2.5, FAISS, embeddings, semantic search, human-in-the-loop systems

ML: Python, PyTorch, TensorFlow, Scikit-learn, Pandas, NLP

Research Methods: held-out evaluation, leakage-free splits, bootstrap confidence intervals, reproducible experiments, experimental design

Research Engineering: Git, DVC, MLflow, Docker, FastAPI, PostgreSQL

PROJECTS

BudgetRoute-LLM - LLM Routing & Evaluation Study

Python | PyTorch | Hugging Face | FAISS | FastAPI | github.com/MedvAx-AI/budgetroute-llm

- **Identified 9.8 pp of routing headroom (65.0% oracle vs 55.2% always-large)** across 1,000 Qwen2.5 runs by comparing five routing strategies on 500 MMLU questions.
- **Reduced p50 inference latency by 15% (390 ms vs 461 ms)** by routing 78% of traffic to Qwen2.5-0.5B and evaluating efficiency trade-offs across small/large models, retrieval, cascade, and abstention.
- **Achieved 1,000/1,000 reproducible answers on a clean rerun** by using leakage-free 300/100/100 splits, pinned model revisions, bootstrap confidence intervals, and held-out evaluation.
- **Strengthened LLM safety and auditability** by adding abstention/human review, traceable RAG evidence, a tamper-evident audit chain, and a PostgreSQL control plane.

Beyond Proximity / SemanticSplat - Research Team Lead

- **Led a 4-person research team through a 150-query evaluation across 5 scenes** by coordinating experimental execution plus ScanNet and Replica pilot studies.

Yacht Resistance MLOps Pipeline

Scikit-learn | Pandas | DVC | MLflow | Docker | github.com/MedvAx-AI/pmldl-yacht-mlops

- **Achieved R2 0.993 and RMSE 1.28 on 70 observations from 5 unseen hulls** by training an Extra Trees pipeline with hull-grouped splits and automatic baseline validation.
- **Made experiments reproducible and continuously deployable** by automating DVC -> MLflow -> FastAPI/Streamlit every 5 minutes and validating releases with 10 CI tests plus model SHA-256 and MLflow run-ID checks.

EDUCATION

Innopolis University | BSc Applied Artificial Intelligence | 2024-2028